

Lecture#02

Random Experiment

A random experiment is an experiment in which,

- a) All possible outcomes of the experiment are known in advance.
- b) Any possible performance of the experiment results in an outcome that is not exactly known in advance.
- c) The experiment can be repeated under identical conditions.

Examples

1. Tossing a coin

- Possible outcomes: Head (H) or Tail (T)
- You cannot predict whether you will get H or T, but you know the sample space is {H, T}.

2. Rolling a die

- Possible outcomes: {1, 2, 3, 4, 5, 6}
- The exact number is unknown before rolling.

3. Drawing a card from a deck

- Sample space: 52 cards (Ace, 2, ..., King of four suits)
- You cannot know which card will be drawn.

4. Choosing a student randomly from a class

- Sample space: All students in the class
- The selected student cannot be predicted in advance.

Equally Likely Events

Two or more events are said to be equally likely if they have the same chance of occurrence.

Example-1: In the experiment of testing a coin, if A be the event of getting a head and B be the event of getting a tail then the events A and B are said to be equally likely events since both the event have same chance of occurrence.

Example-2: In the experiment of testing a dice, if A be the event of getting 1 and B be the event of getting 2 then the events A and B are said to be equally likely events since both the event have same chance of occurrence.

De Morgan's Law

De Morgan's Laws connect complements, unions, and intersections:

1. The complement of a union becomes an intersection of complements
 $(A \cup B)^c = A^c \cap B^c$
2. The complement of an intersection becomes a union of complements
 $(A \cap B)^c = A^c \cup B^c$

Example 1 $S = \{1, 2, 3, 4, 5, 6\}$ $A = \{1, 2, 3, 4\}$ $B = \{3, 4, 5, 6\}$ Find $(A \cup B)^c$. Solution $A \cup B = S$, because together they cover all elements. So the complement inside S is empty. $(A \cup B)^c = \emptyset$ Check with De Morgan: $(A \cup B)^c = A^c \cap B^c$ $A^c = \{5, 6\}$ $B^c = \{1, 2\}$ Intersection is empty. Everything matches.	Example 2 $S = \{a, b, c, d, e\}$ $A = \{a, b, c\}$ $B = \{b, d\}$ Find $(A \cap B)^c$. Solution $A \cap B = \{b\}$ So $(A \cap B)^c = \{a, c, d, e\}$ Check with De Morgan: $(A \cap B)^c = A^c \cup B^c$ $A^c = \{d, e\}$ $B^c = \{a, c, e\}$ Their union is $\{a, c, d, e\}$.
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Permutation

A **permutation** is an arrangement of objects **in a specific order**. Order matters in permutations.

Formula:

If you have **n** objects and want to arrange **r** of them: $n_{Pr} = \frac{n!}{(n-r)!}$, where $n! = n \times (n-1) \times \dots \times 2 \times 1$.

Example 1: Arranging books You have 3 books: A, B, C. How many ways can you arrange all 3 on a shelf? Solution: Here, $n = 3, r = 3$ ${}^3P_3 = \frac{3!}{(3-3)!} = 6$ Arrangements: ABC, ACB, BAC, BCA, CAB, CBA	Example 2: Selecting 2 students from 4 and arranging them Students: A, B, C, D. Select 2 and arrange them. Solution: Let, students: A, B, C, D. $n = 4, r = 2$ ${}^4P_2 = \frac{4!}{(4-2)!} = 12$ Arrangements: AB, AC, AD, BA, BC, BD, CA, CB, CD, DA, DB, DC	Example 3: A club has 5 members. How many ways can they elect a president, vice-president, and secretary (3 distinct positions)? Solution: Positions = 3, Members = 5 ${}^5P_3 = \frac{5!}{(5-3)!} = 60$ So there are 60 ways. Arrangements: ABC, ACB, ABD, ADB, ABE, AEB,...
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Combination

A **combination** is a selection of objects **where order does not matter**.

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Where: n = total objects, r = objects to choose.

Example 1: Choosing books You have 3 books: A, B, C. You want to choose 2 books. ${}^3C_2 = \frac{3!}{2!(3-2)!} = 3$ Selections: AB, AC, BC	Example 2: Choosing students From 4 students: A, B, C, D, select 2. ${}^4C_2 = \frac{4!}{2!(4-2)!} = 6$ Selections: AB, AC, AD, BC, BD, CD	Example 3: A committee of 3 is to be chosen from 5 members. How many ways? ${}^5C_3 = \frac{5!}{3!(5-3)!} = 10$
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Problem 1

A total of 20 people are available: 8 men and 12 women. A committee of 5 members is to be formed such that **3 men** and **2 women** are selected. How many different ways can this committee be formed?

Solution

We need to **choose 3 men from 8** and **2 women from 12**.

Using combinations: Ways to choose men = $8C_3 = \frac{8!}{3!(8-3)!} = 56$

Ways to choose women = $12C_2 = \frac{12!}{2!(12-2)!} = 66$. Since the choices are independent, multiply them:

Total ways = $56 \times 66 = 3696$.

Problem 2

A group has 6 men and 4 women. A committee of 5 is to be formed with at least 1 woman. How many committees?

Solution

Total people = 6 men + 4 women = 10.

Total ways to choose any 5 from 10: $10C_5 = \frac{10!}{5!(10-5)!} = 252$.

Count committees with **no woman** (all men): choose 5 men from 6: $6C_5 = \frac{6!}{5!(6-5)!} = 6$

So committees with **at least one woman** = total – no-woman cases: $252 - 6 = 246$.

Problem 3 A group has 6 men and 4 women. A committee of 5 is to be formed with at least 1 woman and 2 men. How many committees?

Solution: We need 5-person committees with at least 1 woman and at least 2 men. Possible (men, women) splits are:

$$2 \text{ men, } 3 \text{ women: } 6C_2 \times 4C_3 = 15 \times 4 = 60$$

$$3 \text{ men, } 2 \text{ women: } 6C_3 \times 4C_2 = 20 \times 6 = 120$$

$$4 \text{ men, } 1 \text{ women: } 6C_4 \times 4C_1 = 15 \times 4 = 60$$

$$\text{Total} = 60 + 120 + 60 = 240 \text{ committees.}$$

Problem 4

From a standard 52-card deck, 5 cards are drawn at random. What is the number of 5-card hands that contain **exactly one ace**?

Solution:

Choose 1 ace from the 4 aces, and 4 non-aces from the 48 non-aces):

$$4_{C_1} \times 48_{C_4} = 4 \times 194580 = 778320$$

Approaches to compute probability

1. *Classical approach*
2. *Relative frequency approach*
3. *Subjective approach*

Classical approach: This approach works when all outcomes are equally likely. We count the number of favorable outcomes and divide by the number of possible outcomes. It is written as follows:

$$P(A) = \frac{n(A)}{n(S)}$$

Where, $n(A)$ is the number of favorable outcomes that belong to event A , and $n(S)$ is the total number of possible outcomes in the sample space S .

Example

Roll a die.

$$S = \{1,2,3,4,5,6\}, n(S) = 6$$

$$A = \{4\}, n(A) = 1$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{6}$$

Problem 1. You draw a card from 52 cards. What's the probability it is a heart?

Solution

$$S = \{\text{All possible cards}\}, n(S) = 52$$

$$A = \{\text{All hearts}\}, n(A) = 13$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{13}{52} = \frac{1}{4}$$

Problem 2. A bag contains 4 white and 6 red balls. A ball is drawn at random from the bag. What is the probability that is red? What is the probability that is white? Are the events obtaining a red ball and obtaining a white ball equally likely?

Solution:

Here, $S = \{4 \text{ white balls and } 6 \text{ red balls}\}$, $n(S) = 10$.

$R = \{6 \text{ red balls}\}$, $n(R) = 6$; and $W = \{4 \text{ white balls}\}$, $n(W) = 4$

(i) Probability of drawing a red ball,

$$P(R) = \frac{n(R)}{n(S)} = \frac{6}{10} = \frac{3}{5}$$

(ii) Probability of drawing a white ball,

$$P(W) = \frac{n(W)}{n(S)} = \frac{4}{10} = \frac{2}{5}$$

(iii) Since $P(R) \neq P(W)$, the occurrence of the events R and W are not equally likely.

Problem 3. A die is thrown once. Find the following probabilities:

- i. The number obtained is even.
- ii. The number obtained is greater than 4.
- iii. The number obtained is an odd number less than 4.
- iv. Are the events “even number” and “greater than 4” equally likely?

Solution:

The sample space is, $S = \{1, 2, 3, 4, 5, 6\}$. So $n(S) = 6$.

i. Probability of an even number

$A = \{\text{Even outcomes}\} = \{2, 4, 6\}$, so $n(A) = 3$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

ii. Probability of a number greater than 4

$B = \{\text{Numbers greater than } 4\} = \{5, 6\}$, so $n(B) = 2$

$$P(B) = \frac{n(B)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

iii. Probability of an odd number less than 4

$C = \{\text{Odd numbers less than 4}\} = \{1, 3\}$, so $n(C) = 2$

$$P(C) = \frac{n(C)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

iv. Are “even number” and “odd number” equally likely?

Even number probability, $P(A) = \frac{1}{2}$

Odd number probability,

$$P(D) = \frac{n(D)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

Since these are equal. So the events are **equally likely**.

Problem 4 A married couple plans to have two children. Each child is equally likely to be a boy (B) or a girl (G). Find the probabilities of the following events:

- a) both children are boys
- b) both children are girls
- c) at least one girl
- d) at most one boy
- e) exactly one girl
- f) exactly one boy
- g) both are either boys or girls

Solution

Build the sample space (order: first child, second child):

$S = \{BB, BG, GB, GG\}$. So $n(S) = 4$.

a) **Both boys:** $A = \{BB\}$, $n(A) = 1$. $P(A) = \frac{1}{4}$

b) **Both girls:** $A = \{GG\}$, $n(A) = 1$. $P(A) = \frac{1}{4}$

c) **At least one girl:** $A = \{\text{outcomes with at least one G}\} = \{BG, GB, GG\}$. So $n(A) = 3$.

$$P(A) = \frac{3}{4}$$

d) **At most one boy** (0 or 1 boy): $A = \{GG, BG, GB\}$. So, $n(A) = 3$. $P(A) = \frac{3}{4}$

e) **Exactly one girl**: $A = \{BG, GB\}$. So, $n(A) = 2$. $P(A) = \frac{2}{4} = \frac{1}{2}$

f) **Exactly one boy**: $A = \{BG, GB\}$ So, $n(A) = 2$. $P(A) = \frac{2}{4} = \frac{1}{2}$

g) **Both same sex** (both boys or both girls): $A = \{BB, GG\}$. So, $n(A) = 2$. $P(A) = \frac{2}{4} = \frac{1}{2}$

Multiplication Rule

Use this when you want the probability that **A and B** both happen.

1. If A and B are independent

One event does not affect the other.

$$P(A \cap B) = P(A) \times P(B)$$

Example

Flip a coin and roll a die.

Event $A = \{\text{heads}\}$, $P(A) = 1/2$

Event $B = \{\text{getting a 4}\}$, $P(B) = 1/6$

$$P(A \cap B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

Or,

$S = \{(H,1), (H,2), (H,3), (H,4), (H,5), (H,6), (T,1), (T,2), (T,3), (T,4), (T,5), (T,6)\}$, $n(S) = 12$

$C = \{\text{Heads \& four number}\}$, $n(C) = 1$

$$P(A \cap B) = \frac{n(C)}{n(S)} = \frac{1}{12}$$

2. If A and B are dependent

One event affects the other.

$$P(A \cap B) = P(A) \times P(B | A)$$

Example

A bag has 3 red and 2 blue balls. You draw one without replacement.

Event A = first ball is red, $P(A) = \frac{3}{5}$

Given A happened, remaining balls are 2 red, 2 blue.

Event B = second ball is red, $P(B | A) = \frac{2}{4} = \frac{1}{2}$

$$P(A \cap B) = P(A) \times P(B | A) = \frac{3}{5} \times \frac{1}{2} = \frac{3}{10}$$

Or,

$$P(A \cap B) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10}$$

Exercise

1. A fair coin is tossed three times. Compute the probabilities that (a) exactly two tosses result in heads, (b) at most one toss results in heads.
2. A coin is tossed twice and then a die is rolled once.
Find the probability of the following:
 - a) getting exactly one head and an even number on the die
 - b) getting two tails or a number greater than 4 on the die
 - c) getting at least one head and an odd number on the die